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KSK

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:50

Annexure A

Scenario-Based

Code of Conduct:

I Rishykeesh KS (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Rishykeesh KS

90%
KSK

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing $\rightarrow$ /25

School of Electronics $\rightarrow$ /26

School of Mechanical Engineering $\rightarrow$ /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development $\rightarrow$ /24

Quality Assurance $\rightarrow$ /25

Human Resources $\rightarrow$ /26

Executive Management $\rightarrow$ /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The matching routing table entry.
3. The next-hop network/interface used to forward the packet.
4. The default route that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → 192.168.10.0/26

Finance LAN → 192.168.10.64/26

The router interfaces are configured with:

Administration Gateway → 192.168.10.1

Finance Gateway → 192.168.10.65

As the network administrator, perform the following tasks:

1. Determine the network address and broadcast address for each LAN.
2. Identify the valid host IP address range.
3. Suggest suitable IP addresses for three computers in each LAN.
4. Identify the default gateway that hosts in each LAN should use for communication.

	Understanding of Scenario (20%)	Identification of Key Issue (20%)	Application of Concepts (25%)	Decision Making (20%)	Clarity of Response (15%)	
Q1	2	2	2.5	2	1.5	10
Q2	2	1.5	2.5	2	1	9
Q3	1.5	2	2	2	1	8.5
Q4	2	2	2	1.5	1.5	9
Q5	2	1.5	2.5	1.5	1	8.5

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RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context(2)	Shows good understanding with minor gaps(1.5)	Partial understanding; misses key aspects(1)	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved(2)	Identifies most key issues with minor omissions(1.5)	Identifies limited issues; some irrelevant points(1)	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario(2.5)	Applies concepts with minor errors(2)	Limited or partially correct application(1.5)	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale(2)	Makes reasonable decisions with some justification(1.5)	Makes weak decisions with limited justification(1)	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented(1.5)	Generally clear with minor issues(1)	Somewhat unclear or poorly structured(0.5)	Disorganized and difficult to understand

CSA07 – COMPUTER NETWORKS

CO2 ASSESSMENT TOOL 2

SCENARIO-BASED QUESTIONS

Student Detail	Information
Name	Rishiykesh
Register No.	192511466
Course	CSA07 – Computer Networks
CO2	AT2 IP addressing, subnetting and basic routing

QUESTION 1

A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

- School of Computing → /25,
- School of Electronics → /26,
- School of Mechanical Engineering → /27.

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

1. Given Data

Department	Prefix
Computing	/25
Electronics	/26
Mechanical	/27

Main network = 192.168.100.0/24

2. Scenario Understanding

The /24 network must be divided into three non-overlapping subnets using the exact prefixes given.

3. Key Issues

- Find network address and broadcast address.
- Find valid host range.
- Calculate usable host count.
- Find remaining unused addresses.

4. Formula

$$\text{Host bits } H = 32 - \text{prefix length}$$

$$\text{Usable hosts} = 2^H - 2$$

5. Step-by-Step Calculations

Step 1 – Computing /25

$$H = 32 - 25 = 7$$

$$\text{Total addresses} = 2^7 = 128$$

$$\text{Usable hosts} = 128 - 2 = 126$$

Starting network = 192.168.100.0. A 128-address block ends at .127.

Network = .0 | Hosts = .1 to .126 | Broadcast = .127

Step 2 – Electronics /26

$$\text{Next network} = .127 + 1 = 192.168.100.128$$

$$H = 32 - 26 = 6$$

$$\text{Total addresses} = 2^6 = 64$$

$$\text{Usable hosts} = 64 - 2 = 62$$

Block = .128 to .191. Therefore hosts = .129 to .190 and broadcast = .191.

Step 3 – Mechanical /27

$$\text{Next network} = .191 + 1 = 192.168.100.192$$

$$H = 32 - 27 = 5$$

$$\text{Total addresses} = 2^5 = 32$$

$$\text{Usable hosts} = 32 - 2 = 30$$

Block = .192 to .223. Therefore hosts = .193 to .222 and broadcast = .223.

Step 4 – Unused range

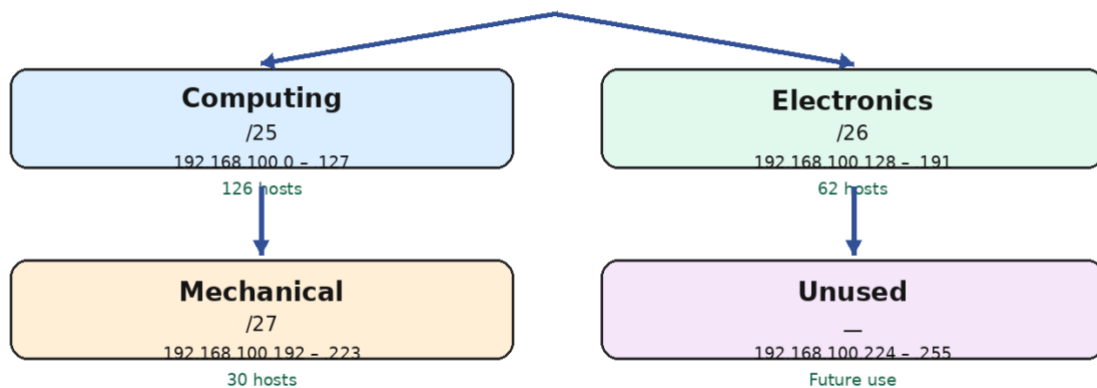
$$\text{First unused address} = .223 + 1 = .224$$

The original /24 ends at .255.

$$\text{Unused} = 192.168.100.224 - 192.168.100.255$$

Smart Campus Subnetting

192.168.100.0/24



6. Final Answer

Department	Network	Host Range	Broadcast	Usable
Computing	192.168.100.0/25	.1–.126	.127	126
Electronics	192.168.100.128/26	.129–.190	.191	62
Mechanical	192.168.100.192/27	.193–.222	.223	30

Decision / Rationale: The allocation is non-overlapping and the unused .224–.255 range can be reserved for future expansion.

Interpretation: The subnetting gives each department a separate address range. The remaining addresses can be reserved for future Smart Campus expansion.

QUESTION 2

A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

- Software Development → /24,
- Quality Assurance → /25,
- Human Resources → /26,
- Executive Management → /27.

As the network administrator, perform the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid range of host IP addresses.
3. Calculate the number of usable hosts in each subnet.
4. Identify the unused IP address range remaining within the allocated IP block.

1. Given Data

Department	Prefix
Software Development	/24
Quality Assurance	/25
Human Resources	/26
Executive Management	/27

Main network = 10.10.0.0/16

2. Scenario Understanding

A large /16 block is divided sequentially into four department networks. The prefixes determine the block sizes.

3. Key Issues

- Calculate host capacity for each prefix.
- Find each next network boundary.
- Find host range and broadcast.
- Find the remaining /16 address space.

4. Formula

$$\text{Usable hosts} = 2^{(32 - \text{prefix})} - 2$$

5. Step-by-Step Calculations

Step 1 – Software Development /24

$$H = 32 - 24 = 8$$

$$\text{Total} = 2^8 = 256; \text{ Usable} = 256 - 2 = 254$$

Network = 10.10.0.0; hosts = 10.10.0.1–254; broadcast = 10.10.0.255.

Step 2 – Quality Assurance /25

$$\text{Next network} = 10.10.0.255 + 1 = 10.10.1.0$$

$$H = 32 - 25 = 7$$

$$\text{Total} = 2^7 = 128; \text{ Usable} = 128 - 2 = 126$$

Network = 10.10.1.0; hosts = .1–.126; broadcast = .127.

Step 3 – Human Resources /26

$$\text{Next network} = 10.10.1.127 + 1 = 10.10.1.128$$

$$H = 32 - 26 = 6$$

$$\text{Total} = 2^6 = 64; \text{ Usable} = 64 - 2 = 62$$

Network = 10.10.1.128; hosts = .129–.190; broadcast = .191.

Step 4 – Executive Management /27

$$\text{Next network} = 10.10.1.191 + 1 = 10.10.1.192$$

$$H = 32 - 27 = 5$$

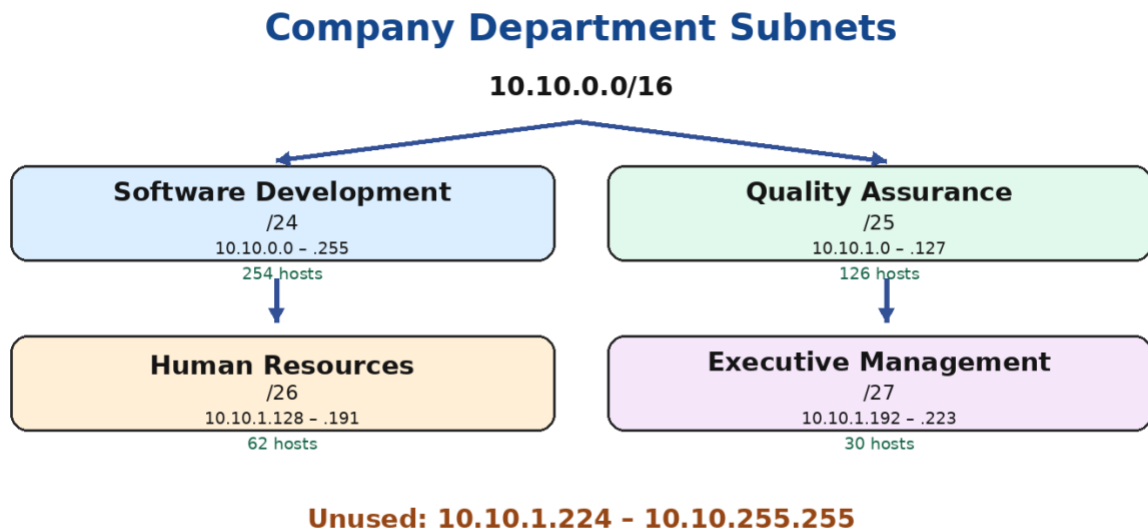
$$\text{Total} = 2^5 = 32; \text{ Usable} = 32 - 2 = 30$$

Network = 10.10.1.192; hosts = .193–.222; broadcast = .223.

Step 5 – Unused range

$$\text{First unused} = 10.10.1.223 + 1 = 10.10.1.224$$

$$\text{Unused} = 10.10.1.224-10.10.255.255$$



6. Final Answer

Department	Network	Host Range	Broadcast	Usable
Development	10.10.0.0/24	.1-.254	.255	254
QA	10.10.1.0/25	.1-.126	.127	126
HR	10.10.1.128/26	.129-.190	.191	62
Executive	10.10.1.192/27	.193-.222	.223	30

Decision / Rationale: The subnet sizes match the given prefixes and the remaining /16 space is available for future departments.

Interpretation: The larger departments receive larger subnet blocks, while the unused part of the /16 network remains available for future departments and services.

QUESTION 3

A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network: 2001:0db8:abcd:0000:0000:0000:0000:0000/64.

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its compressed notation.
2. Expand the compressed address back into its full hexadecimal representation.
3. Identify the network prefix using the /64 prefix length.
4. Calculate the total number of host addresses supported within the subnet.
5. Determine the total number of possible host addresses available within this network.

1. Given Data

Parameter	Value
IPv6 network	2001:0db8:abcd:0000:0000:0000:0000:0000/64
Prefix	/64
Address size	128 bits

2. Scenario Understanding

The hospital needs a large IPv6 address space. The /64 prefix divides the 128-bit address into a 64-bit network part and a 64-bit host/interface part.

3. Key Issues

- Compress the address correctly.
- Expand it without changing its value.
- Identify the first 64 bits as the network prefix.
- Calculate the 64-bit host space.

4. Step-by-Step Calculations

Step 1 – Write the eight IPv6 groups

2001 : 0db8 : abcd : 0000 : 0000 : 0000 : 0000 : 0000

Step 2 – Remove leading zeros inside each group

2001 : db8 : abcd : 0 : 0 : 0 : 0 : 0

Step 3 – Replace the longest consecutive zero groups with ::

2001:db8:abcd::/64

Step 4 – Expand the compressed form

The :: represents the required consecutive zero groups so that there are exactly eight groups.

2001:db8:abcd::/64

= 2001:0db8:abcd:0000:0000:0000:0000:0000/64

Step 5 – Identify the /64 network prefix

First 64 bits = first 4 groups

Network prefix = 2001:0db8:abcd:0000/64

Equivalent compressed prefix: 2001:db8:abcd::/64.

Step 6 – Calculate host bits

$$\text{Host bits} = 128 - 64 = 64$$

Step 7 – Calculate total address capacity

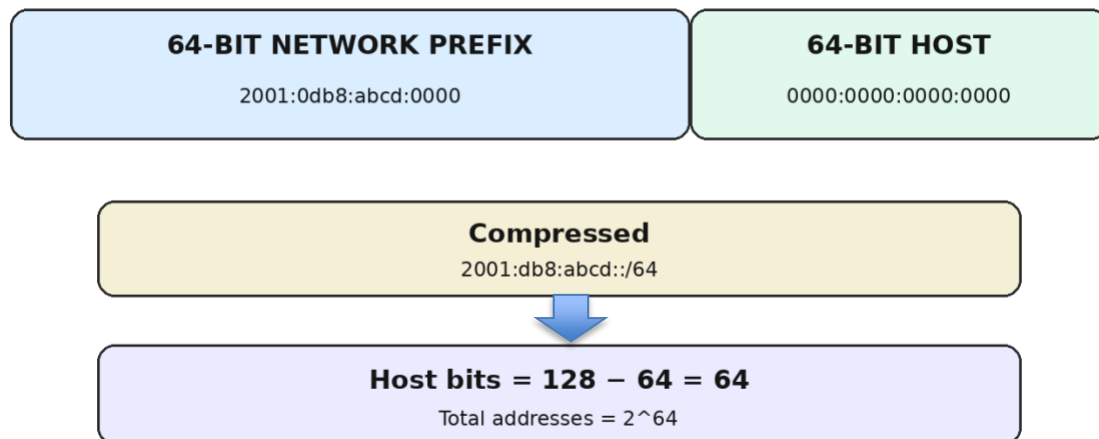
$$\begin{aligned}\text{Total addresses} &= 2^{64} \\ &= 18,446,744,073,709,551,616\end{aligned}$$

Step 8 – IPv6 addressing note

IPv6 does not use an IPv4-style broadcast address. The calculated 2^{64} is the address space of the /64 prefix.

IPv6 /64 Structure

2001:0db8:abcd:0000:0000:0000:0000:0000 /64



5. Final Answer

Item	Answer
Compressed	2001:db8:abcd::/64
Expanded	2001:0db8:abcd:0000:0000:0000:0000:0000/64
Network prefix	2001:db8:abcd::/64
Host bits	64
Total addresses	$2^{64} = 18,446,744,073,709,551,616$

Decision / Rationale: The /64 network provides a very large address space suitable for the hospital's connected devices and future IoT expansion.

Interpretation: A /64 IPv6 prefix leaves 64 host bits, providing a very large address space suitable for large numbers of hospital devices and IoT nodes.

QUESTION 4

An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries: 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24.

A packet arrives at the router with the destination IP address: 192.168.2.45.

As the routing engineer, determine:

1. The destination subnet to which the packet belongs.
2. The matching routing table entry.
3. The next-hop network/interface used to forward the packet.
4. The default route that should be used if the destination does not match any routing table entry.

1. Given Data

Routing Entry
192.168.1.0/24
192.168.2.0/24
192.168.3.0/24

Destination IP = 192.168.2.45

2. Scenario Understanding

The router must compare the destination address with its routing table and select the matching network.

3. Key Issues

- Understand the /24 network boundary.
- Find the destination subnet.
- Find the matching route.
- State the forwarding interface and default route.

4. Step-by-Step Reasoning

Step 1 – Convert /24 to subnet mask

/24 = 255.255.255.0

Therefore, the first three octets identify the network.

Step 2 – Separate network and host

Destination = 192.168.2.45

Network part = 192.168.2

Host part = 45

So the destination network is 192.168.2.0/24.

Step 3 – Compare with routing table

- 192.168.1.0/24 → does not match.
- 192.168.2.0/24 → MATCH.
- 192.168.3.0/24 → does not match.

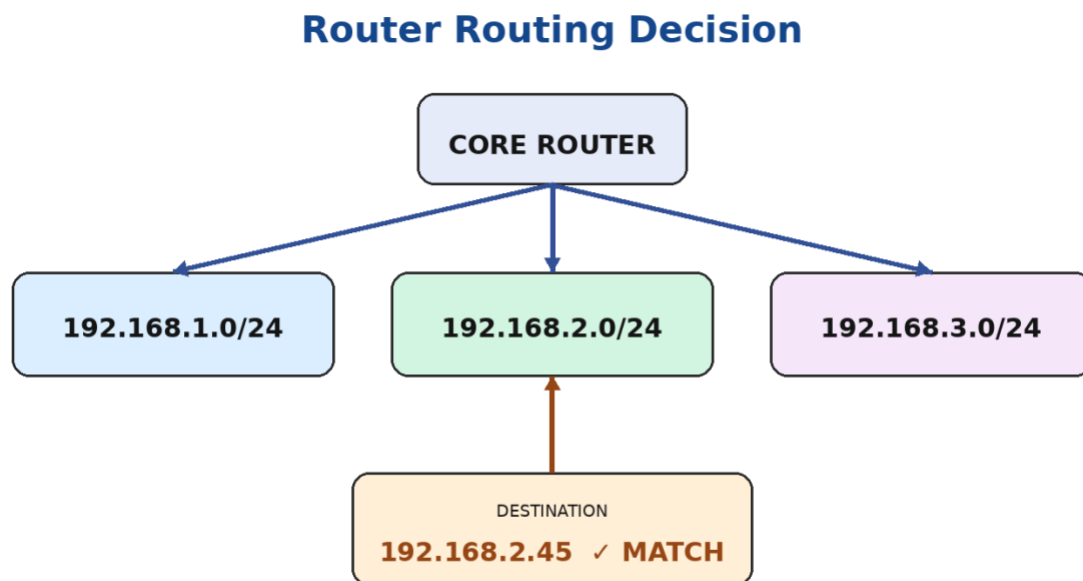
Step 4 – Forward the packet

The router forwards the packet through the interface connected to 192.168.2.0/24. The question does not provide a specific interface name or next-hop IP, so none should be invented.

Step 5 – Default route

Default IPv4 route = 0.0.0.0/0

If no specific route matches, the configured default route is considered.



5. Final Answer

Requirement	Answer
Destination subnet	192.168.2.0/24
Matching route	192.168.2.0/24
Forwarding interface	Interface connected to 192.168.2.0/24
Default route	0.0.0.0/0

Decision / Rationale: The specific 192.168.2.0/24 route is selected because it directly matches the destination. The default route is only used when no suitable specific route exists.

Interpretation: The destination 192.168.2.45 belongs to 192.168.2.0/24, so the router selects that specific route instead of the default route.

QUESTION 5

A company establishes two office departments connected through a Layer-3 router.

The allocated subnet information is as follows:

- Administration LAN → 192.168.10.0/26;
- Finance LAN → 192.168.10.64/26.

The router interfaces are configured with:

- Administration Gateway → 192.168.10.1
- Finance Gateway → 192.168.10.65.

As the network administrator, perform the following tasks:

- Determine the network address and broadcast address for each LAN.
- Identify the valid host IP address range.
- Suggest suitable IP addresses for three computers in each LAN.
- Identify the default gateway that hosts in each LAN should use for communication.

1. Given Data

LAN	Subnet	Gateway
Administration	192.168.10.0/26	192.168.10.1
Finance	192.168.10.64/26	192.168.10.65

2. Scenario Understanding

There are two separate /26 LANs. A Layer-3 router connects them. Each LAN must use the router interface in that same subnet as its default gateway.

3. Key Issues

- Find network and broadcast addresses.
- Find valid host ranges.
- Choose three valid computer addresses.
- Use the correct gateway for each LAN.

4. Step-by-Step Calculations

Step 1 – Find the subnet mask

$$\text{/26} = 255.255.255.192$$

Step 2 – Find block size

$$\text{Block size} = 256 - 192 = 64$$

Therefore the /26 ranges are 0–63, 64–127, 128–191, and so on.

Step 3 – Administration LAN

$$\text{Network} = 192.168.10.0$$

$$\text{Broadcast} = 0 + 63 = 192.168.10.63$$

$$\text{Valid hosts} = 192.168.10.1\text{--}192.168.10.62$$

Gateway .1 is already assigned. Choose PC addresses .2, .3, and .4.

Step 4 – Finance LAN

Network = 192.168.10.64

Broadcast = 64 + 63 = 192.168.10.127

Valid hosts = 192.168.10.65–192.168.10.126

Gateway .65 is already assigned. Choose PC addresses .66, .67, and .68.

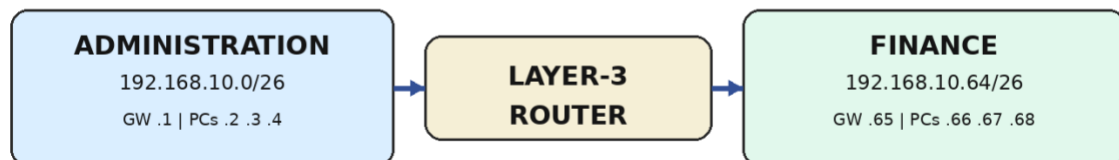
Step 5 – Verify addresses

- Administration: .0 is network, .63 is broadcast, .1 is gateway; .2/.3/.4 are valid hosts.
- Finance: .64 is network, .127 is broadcast, .65 is gateway; .66/.67/.68 are valid hosts.

Step 6 – Communication through Layer-3 router

A host sends traffic outside its own subnet to its local gateway. The router then forwards it toward the other subnet.

Two LANs via Layer-3 Router



Broadcast: .63

Broadcast: .127

Host → Local Gateway → Router → Other LAN

Different /26 networks communicate through Layer-3 routing

5. Final Answer

LAN	Network	Host Range	Broadcast	Gateway	3 PCs
Administration	192.168.10.0/26	.1–.62	.63	.1	.2, .3, .4
Finance	192.168.10.64/26	.65–.126	.127	.65	.66, .67, .68

Decision / Rationale: Each LAN uses the gateway configured inside its own subnet. The selected PC addresses avoid the network, broadcast, and gateway addresses.

Interpretation: Each LAN has its own /26 subnet and gateway. A device sends traffic to the other LAN through its local gateway and the Layer-3 router.